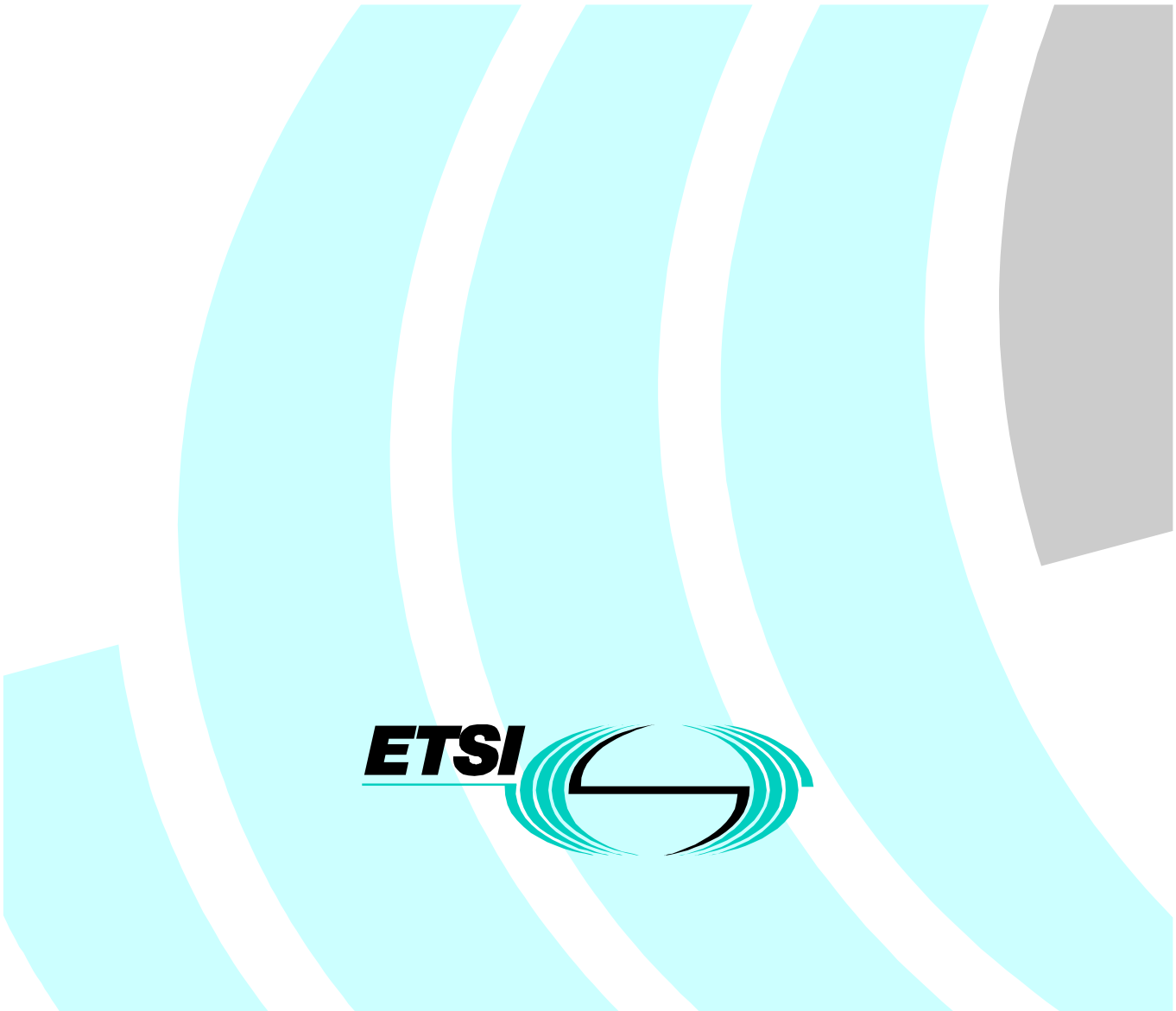
EN 300 443-1 V1.3.5 (1998-10)

European Standard (Telecommunications series)

Broadband Integrated Services Digital Network (B-ISDN); Digital Subscriber Signalling System No. two (DSS2) protocol; B-ISDN user-network interface layer 3 specification for basic call/bearer control; Part 1: Protocol specification

[ITU-T Recommendation Q.2931 (1995), modified]



Reference
REN/SPS-05136-1 (43o90j3c.PDF)

Keywords
B-ISDN, DSS2, UNI, layer 3, basic

ETSI

Postal address
F-06921 Sophia Antipolis Cedex - FRANCE

Office address
650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16
Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Internet
secretariat@etsi.fr
http://www.etsi.org

Copyright Notification

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 1998.
All rights reserved.

Intellectual Property Rights

IPRs essential or potentially essential to the present document may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available **free of charge** from the ETSI Secretariat. Latest updates are available on the ETSI Web server (http://www.etsi.org/ipr).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Signalling Protocols and Switching (SPS).

The present document is part 1 of a multi-part standard covering the specification of the Broadband Integrated Services Digital Network (B-ISDN) user-network interface layer 3 protocol for basic call/bearer control of Digital Subscriber Signalling System No. two (DSS2) as described below:

Part 1: "Protocol specification [ITU-T Recommendation Q.2931 (1995), modified]";

Part 2: "Protocol Implementation Conformance Statement (PICS) proforma specification".

Part 3: "Test Suite Structure and Test Purposes (TSS&TP) specification for the user";

Part 4: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the user";

Part 5: "TSS&TP specification for the network";

Part 6: "ATS and partial PIXIT proforma specification for the network".

National transposition dates	
Date of adoption of this EN:	30 October 1998
Date of latest announcement of this EN (doa):	31 January 1999
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	31 July 1999
Date of withdrawal of any conflicting National Standard (dow):	31 July 1999

Endorsement notice

The elements of ITU-T Recommendation Q.2931 (1995), apply, with the following modifications:

NOTE: New or modified text is indicated using sidebars. In addition, underlining and/or strike-out are used to highlight detailed modifications where necessary.

Page 1, clause 1

Replace clause 1 (excluding subclauses 1.1 to 1.3) by:

1 Scope

The present document specifies the stage three of on-demand connection-oriented release 1 basic telecommunication services for the pan-European Broadband Integrated Services Digital Network (B-ISDN) as provided at the T_B reference point or coincident S_B and T_B reference point (as defined in ITU-T Recommendation I.413 [19]) by means of the Digital Subscriber Signalling System No. two (DSS2) protocol. Stage three identifies the protocol procedures and switching functions needed to support a telecommunications service (see CCITT Recommendation I.130 [59]).

In addition, the present document specifies the protocol requirements at the T_B reference point where the service is provided to the user via a private B-ISDN.

NOTE 1: Procedures at the T_B reference point, to support the access of a private B-ISDN to the public B-ISDN, are **not** explicitly identified in the present document, however, some procedures are applicable only at the T_B reference point.

The present document also specifies the particular features required to provide 64 kbit/s based circuit-mode ISDN services in B-ISDN and to support access interworking between B-ISDN and N-ISDN.

A basic telecommunications service is a fundamental type of service. It forms the basis on which supplementary services may be added.

NOTE 2: Specific requirements of individual B-ISDN connection-oriented basic telecommunication services are not covered by the present document.

Further parts of the present document provide the method of testing and detailed application specific requirements to determine conformance to the present document.

The present document is applicable to equipment supporting connection-oriented basic B-ISDN telecommunication services, to be attached at either side of a T_B reference point or coincident S_B and T_B reference points when used as an access to the public B-ISDN.

Page 1, subclause 1.1

Replace the text of subclause 1.1 by:

1.1.1 Definitions and abbreviations

Definitions and abbreviations are provided in clauses J.1 and J.2, respectively.

1.1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.

- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[59] CCITT Recommendation I.130 (1988): "Method for the characterization of telecommunication services supported by an ISDN and network capabilities of an ISDN".

[60] ETS 300 402-2: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Data link layer; Part 2: General protocol specification [ITU-T Recommendation Q.921 (1993), modified]".

[61] EN 300 403-1: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit-mode basic call control; Part 1: Protocol specification [ITU-T Recommendation Q.931 (1993), modified]".

[62] ETS 300 436-1: "Broadband Integrated Services Digital Network (B-ISDN); Signalling ATM Adaptation Layer (SAAL); Service Specific Connection Oriented Protocol (SSCOP); Part 1: Protocol specification [ITU-T Recommendation Q.2110 (1995), modified]".

[63] ETS 300 437-1: "Broadband Integrated Services Digital Network (B-ISDN); Signalling ATM Adaptation Layer (SAAL); Service Specific Co-ordination Function (SSCF) for support of signalling at the User-Network Interface (UNI); Part 1: Specification of SSCF at UNI [ITU-T Recommendation Q.2130 (1995), modified]".

[64] ETS 300 486-1: "Broadband Integrated Services Digital Network (B-ISDN); Meta-signalling protocol; Part 1: Protocol specification [ITU-T Recommendation Q.2120 (1995), modified]".

[65] EN 301 068-1: "Broadband Integrated Services Digital Network (B-ISDN); Digital Subscriber Signalling System No. two (DSS2) protocol; Connection characteristics; ATM transfer capability and traffic parameter indication; Part 1: Protocol specification [ITU-T Recommendations Q.2961.1 (1995), Q.2961.2 to Q.2961.4 (1997), modified]".

Throughout the text of ITU-T Recommendation Q.2931

Replace references as shown in the following table.

Reference in ITU-T Recommendation Q.2931	Modified reference
ITU-T Recommendation Q.921	ITU-T Recommendation Q.921 as modified by ETS 300 402-2 [60]
ITU-T Recommendation Q.931	ITU-T Recommendation Q.931 as modified by EN 300 403-1 [61]
ITU-T Recommendation Q.2110	ITU-T Recommendation Q.2110 as modified by ETS 300 436-1 [62]
ITU-T Recommendation Q.2120	ITU-T Recommendation Q.2120 as modified by ETS 300 486-1 [64]
ITU-T Recommendation Q.2130	ITU-T Recommendation Q.2130 as modified by ETS 300 437-1 [63]
ITU-T Recommendation Q.2961.2	ITU-T Recommendation Q.2961.2 as modified by EN 301 068-1 [65]

Page 13, table 3-8/Q.2931, note 5

Add the following text to note 5:

The maximum length of this information element is 25 octets.

Page 14, table 3-8/Q.2931, note 7

Add the following text to note 7:

Not included in the network-to-user direction for basic call control, but may be included for some supplementary services. The maximum length of this information element is 26 octets.

Page 14, table 3-8/Q.2931, note 8

Add the following text to note 8:

Not included in the network-to-user direction for basic call control, but may be included for some supplementary services.

Page 14, table 3-8/Q.2931, note 13

Replace note 13 by:

NOTE 13: It is mandatory for the user to include the Broadband sending complete information element when *en bloc* sending procedures are used. On its receipt, the network shall apply the *en bloc* sending procedure.

It is mandatory for the network to include the Broadband sending complete information element. If the Broadband sending complete information element is not included, the called user shall not apply the missing mandatory element error handling procedures.

Page 23, table 3-19/Q.2931, note 2

Add the following text to note 2:

The maximum length of this information element is 25 octets.

Page 23, table 3-19/Q.2931, note 4

Add the following text to note 4:

Not included in the network-to-user direction for basic call control, but may be included for some supplementary services. The maximum length of this information element is 26 octets.

Page 23, table 3-19/Q.2931, note 5

Add the following text to note 5:

Not included in the network-to-user direction for basic call control, but may be included for some supplementary services.

Page 49, subclause 4.5.6, second paragraph

Insert the following note after the second paragraph:

NOTE: When the user requires the use of end-to-end OAM F5 flows, it is recommended that this is declared explicitly within the OAM traffic descriptor information element (see subclause 4.5.24) in order to enable the network to identify the actual user plane information rates. Currently this is needed when the user expects the separate policing and shaping functions (when applied as a network option) of the user plane information flow and of the end-to-end user originated OAM F5 flow to be performed by the network. Other cases may be identified in the future.

Page 54, subclause 4.5.9, third paragraph

Insert after the third paragraph:

Octet groups 5 (layer 1 id), 6 (layer 2 id) and 7 (layer 3 id) of the Broadband low layer information element are not position independent, but if present at all, shall be sent in the order as specified in figure 4-16/Q.2931.

Page 60, subclause 4.5.11, third paragraph

Replace the third paragraph by:

The maximum length of this information element is 25 octets.

Page 61, figure 4-18/Q.2931, note 2

Delete note 2 related to "NSAP address octets". This field is not supported.

Page 62, table 4-12/Q.2931, note 2

Replace note 2 by:

NOTE 2: The type of number "unknown" is used when the user or the network have no knowledge of the type of number (e.g. international number, national number, etc.). In this case, the number digits field is organized according to the network dialling plan; e.g. prefix or escape digits might be present.

Page 62, table 4-12/Q.2931, "Addressing/numbering plan (octet 5)"

Change codepoint "0 0 1 0" to "reserved" and delete the related note 9.

Page 64, subclause 4.5.13, second paragraph

Replace the second sentence by:

The maximum length of this information element is 26 octets.

Page 65, figure 4-20/Q.2931

Delete note 2 related to "NSAP address octets". This field is not supported.

Page 66, table 4-14/Q.2931, note 2

Replace note 2 by:

NOTE 2: The type of number "unknown" is used when the user or the network have no knowledge of the type of number (e.g. international number, national number, etc.). In this case, the number digits field is organized according to the network dialling plan; e.g. prefix or escape digits might be present.

Page 66, table 4-14/Q.2931, "Addressing/numbering plan (octet 5)" and note 9

Change codepoint "0 0 1 0" to "reserved" and delete the related note 9.

Page 77, subclause 4.5.24, first paragraph

Insert the following note after the first paragraph:

NOTE: Future enhancements to the present document may lead to extensions to this information element (or the definition of other means) to allow explicit declaration of end-to-end user originated OAM F5 flow to accommodate other user management requirements.

Page 81, clause 5, note

Delete the note. Annex H does not form part of the present document.

Page 86, subclause 5.2.1, second paragraph

Replace the second sentence by:

In this case, the SETUP message shall contain the Broadband sending complete information element.

Page 86, subclause 5.2.1, fourth paragraph

Replace the fourth paragraph by:

If the SETUP includes the Broadband sending complete information element, the *en bloc* receiving procedure shall be followed. Therefore, on its receipt, the user shall apply the *en bloc* receiving procedure.

Page 110, table 7-3/Q.2931, T301

The implementation of user side timer T301 shall be optional.

Page 118, annex A

Annex A has the status of a normative annex.

The user side SDL extensions for symmetric call operation are not supported.

The network side SDL extensions for symmetric call operation are not supported.

Page 207, annex B

Annex B has the status of a normative annex.

Page 208, annex C

Annex C has the status of a normative annex.

Page 210, annex D

Annex D has the status of a normative annex.

Page 211, annex E

Annex E has the status of a normative annex.

Page 218, annex F

Annex F has the status of a normative annex.

Page 220, annex G

Delete Annex G. It does not form part of the present document.

Page 221, annex H

Delete Annex H. It does not form part of the present document.

Page 221, annex I

Annex I has the status of a normative annex.

Page 222, annex J

Annex J has the status of a normative annex.

Page 222, clause J.1

Add the following note:

NOTE: ITU-T Recommendation Q.2931 frequently uses the term "connection". ETSI has decided to use the term "bearer" instead (i.e. if this term is used in the context of a bearer for a connection-oriented bearer service).

Page 229, annex K

Annex K has the status of a normative annex.

Page 231, annex L

Annex L has the status of an informative annex.

Page 234, appendix I

Appendix I has the status of an informative annex.

Page 236, appendix II

Appendix II has the status of an informative annex.

Page 238, appendix III

Delete appendix III. It does not form part of the present document.

Annex Z.A (informative)
Identification of changes from previous standards

Z.A.1 Identification of changes from ETS 300 443-1 edition 1 to the present document

The latest version of the standard has been produced to incorporate the 1997 amendment to ITU-T Recommendation Q.2931 into the ETSI endorsement. This change provides for the following:

In subclause 4.5.7 (Broadband bearer capability) of ITU-T Recommendation Q.2931 (1995), the contents of the subclause (including Figure 4-14/Q.2931 and Table 4-8/Q.2931) is to be replaced with "see ITU-T Recommendation Q.2961.2".

This amendment to ITU-T Recommendation Q.2931 has been prepared to allow the creation of ITU-T Recommendation Q.2961.2 to contain all the information on the encoding of the Broadband bearer capability information element in a manner that is consistent with the second edition of ITU-T Recommendation I.371. This amendment is issued in conjunction with ITU-T Recommendation I.2961.2, which is designed to be compatible with implementations conforming to the 1st edition of ITU-T Recommendation I.371 and ITU-T Recommendation Q.2931 (1995).

Within ITU-T Recommendation Q.2961.2, the changes from ITU-T Recommendation Q.2931 (1995) are as follows:

- Octet 5a of the Broadband bearer capability information element was restructured to support ATM transfer capabilities (ATCs) specified in the second edition of ITU-T Recommendation I.371. This restructuring is backward compatible with the first edition of ITU-T Recommendation Q.2931 (i.e. values of octet 5a defined in the first edition of ITU-T Recommendation Q.2931 still have the same meaning). These values may be either those used on transmission and reception, those not generated by terminal equipment complying with ITU-T Recommendation Q.2961.2, or those that are reserved.

History

Document history				
Edition 1	April 1996	Publication as ETS 300 443-1		
V1.2.1	December 1997	Public Enquiry	PE 9815:	1997-12-12 to 1998-04-10
V1.3.4	August 1998	Vote	V 9843:	1998-08-25 to 1998-10-23
V1.3.5	October 1998	Publication		

ISBN 2-7437-2634-2
Dépôt légal : Octobre 1998